

Intro to Predictive AI: Logistic Regression

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Predictive AI with health data

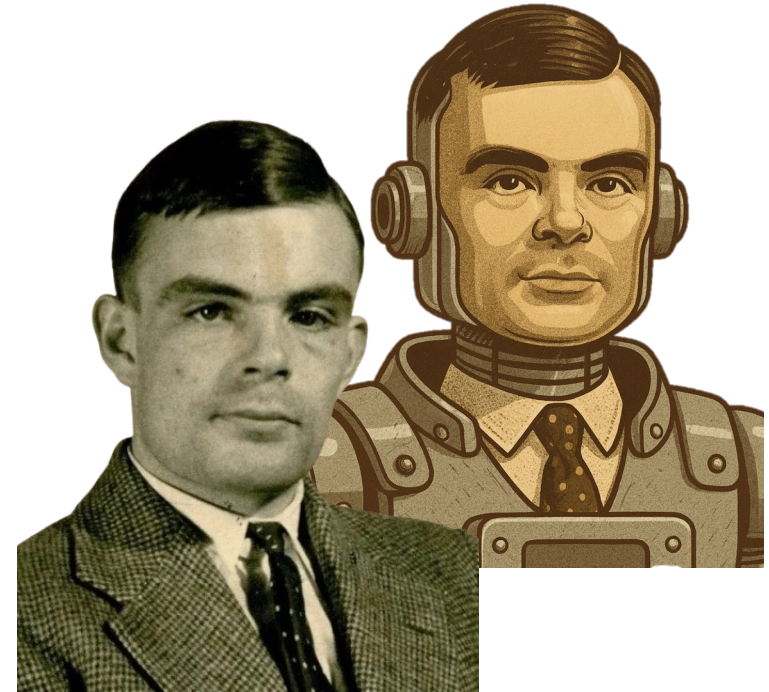
An overview of what we'll be covering

What is AI?

A scientific field concerned with developing **intelligent machines**.

What is **intelligence**?

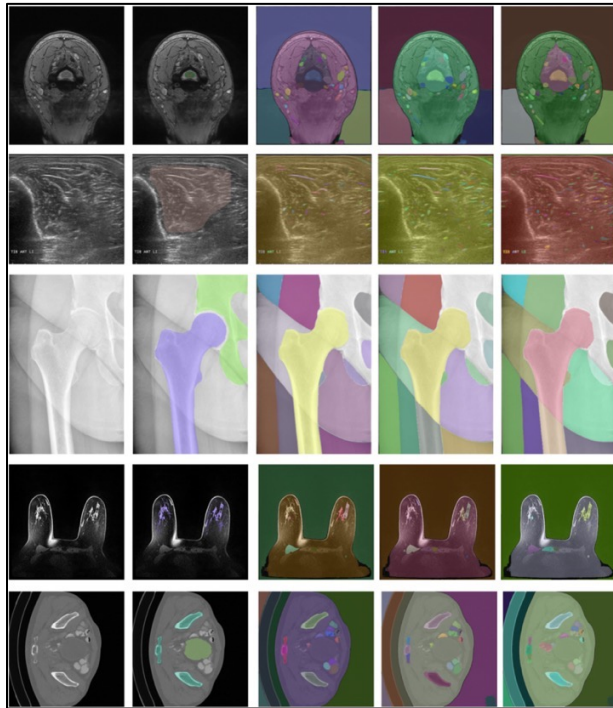
- learn from experience
- solve problems
- make decisions



Turing or TuringBot?

Branches of AI with high clinical impact

Computer vision



Mazurowski, Maciej A., et al. "Segment anything model for medical image analysis: an experimental study." *Medical Image Analysis* 89 (2023): 102918.

Natural language processing

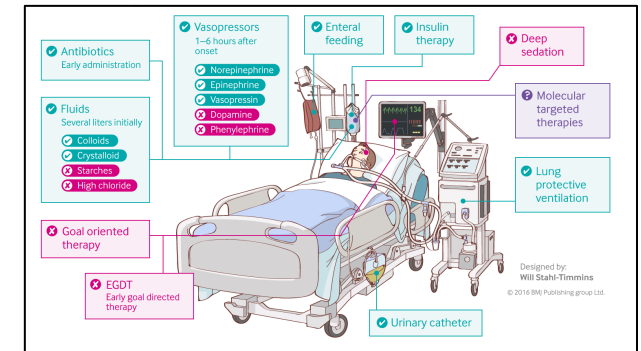


Please provide a very brief, bulleted summary of evidence supporting the effectiveness of the Early Start Denver Model for early autism intervention.

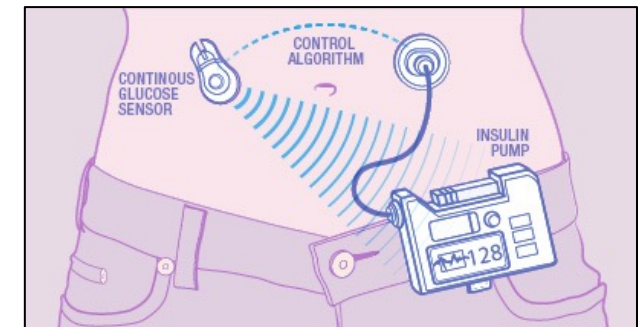
Here's a concise, evidence-focused, bulleted summary:

- **Improved cognitive and language outcomes:** Multiple randomized controlled trials (RCTs) show that children receiving the Early Start Denver Model (ESDM) demonstrate significantly greater gains in IQ, adaptive behavior, and expressive/receptive language compared with control groups receiving standard community services.
- **Enhanced adaptive functioning:** Studies report improvements in social communication, daily living skills, and overall adaptive behaviors, sustained beyond treatment.
- **Reduced autism symptom severity:** Follow-up analyses indicate modest but significant reductions in ASD symptom severity relative to baseline and comparison groups.
- **Neural evidence of typical processing:** EEG studies suggest that children treated with EDSM show more typical patterns of brain activity in response to social stimuli, indicating neural-level changes.
- **Generalization to community settings:** Effectiveness has been demonstrated in both controlled university-based settings and community-based early intervention programs, supporting scalability.
- **Durable effects:** Longitudinal follow-ups show persistence of developmental gains several years post-intervention.

Reinforcement learning

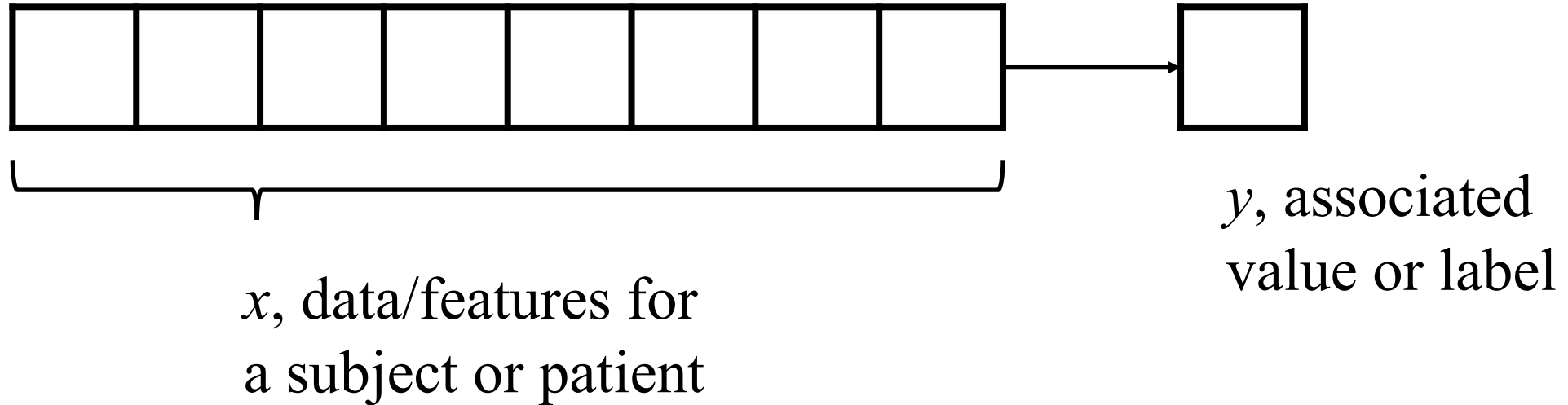


RL-based sepsis management



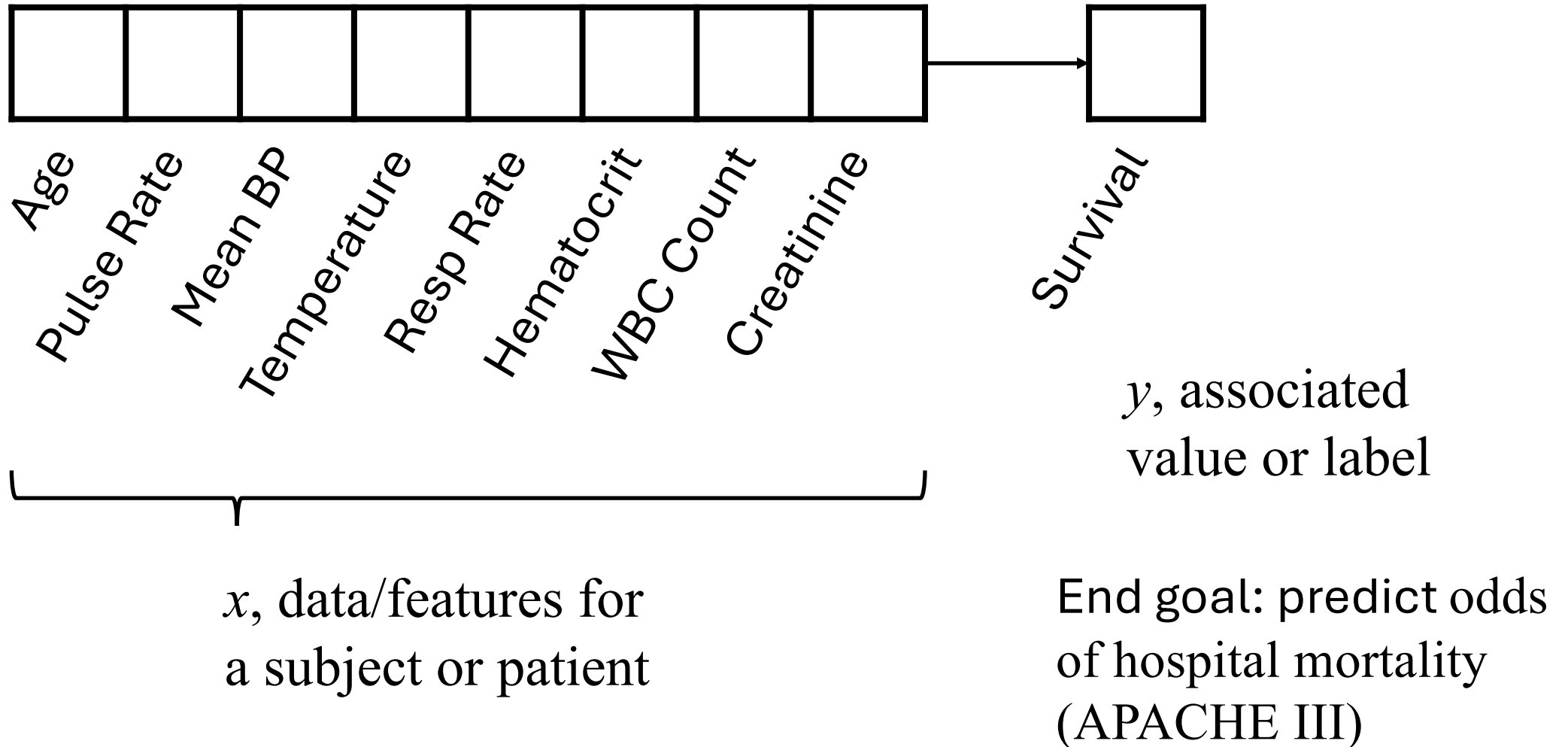
closed-loop BG control

The *machine* in machine learning



End goal: predict y from x

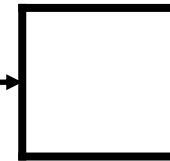
Simple models often work well for clinical data!



Computer vision: prediction via convolutional neural network



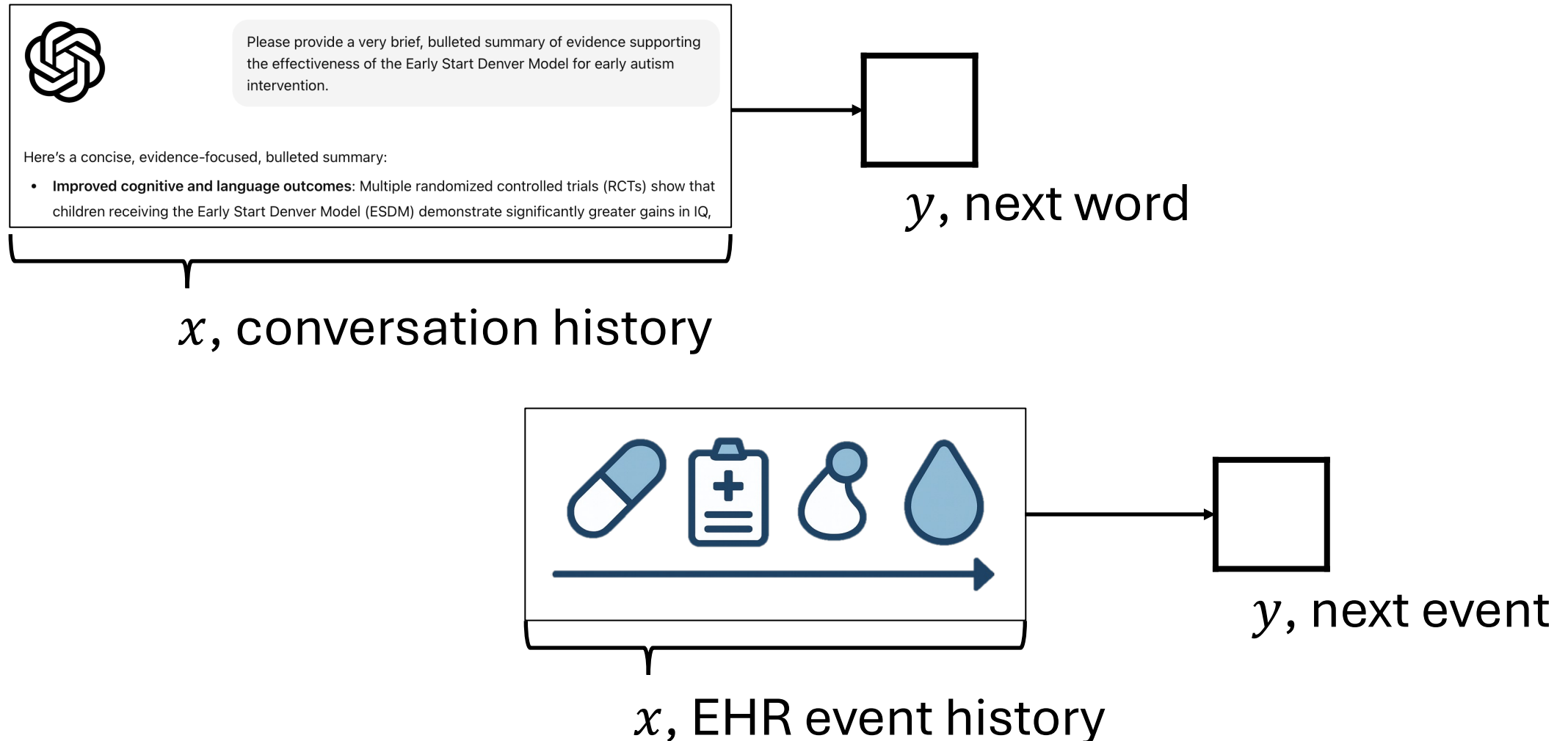
x , retinal image



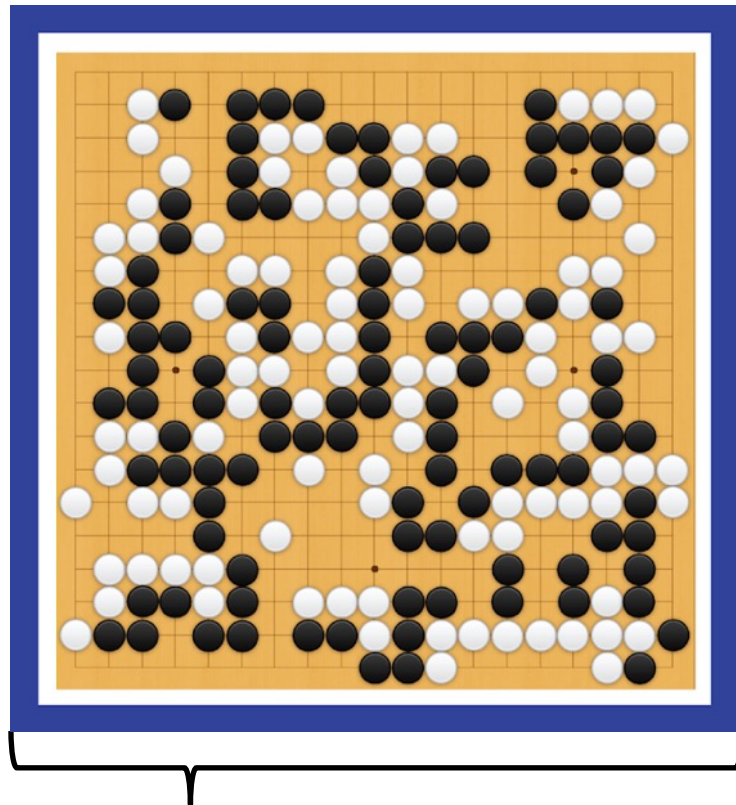
y , referable
diabetic retinopathy

End goal: predict y from x

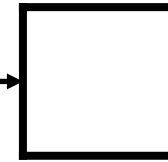
NLP and clinical sequences: prediction via stacked attention networks



Reinforcement Learning: state to action predictions via dense NN



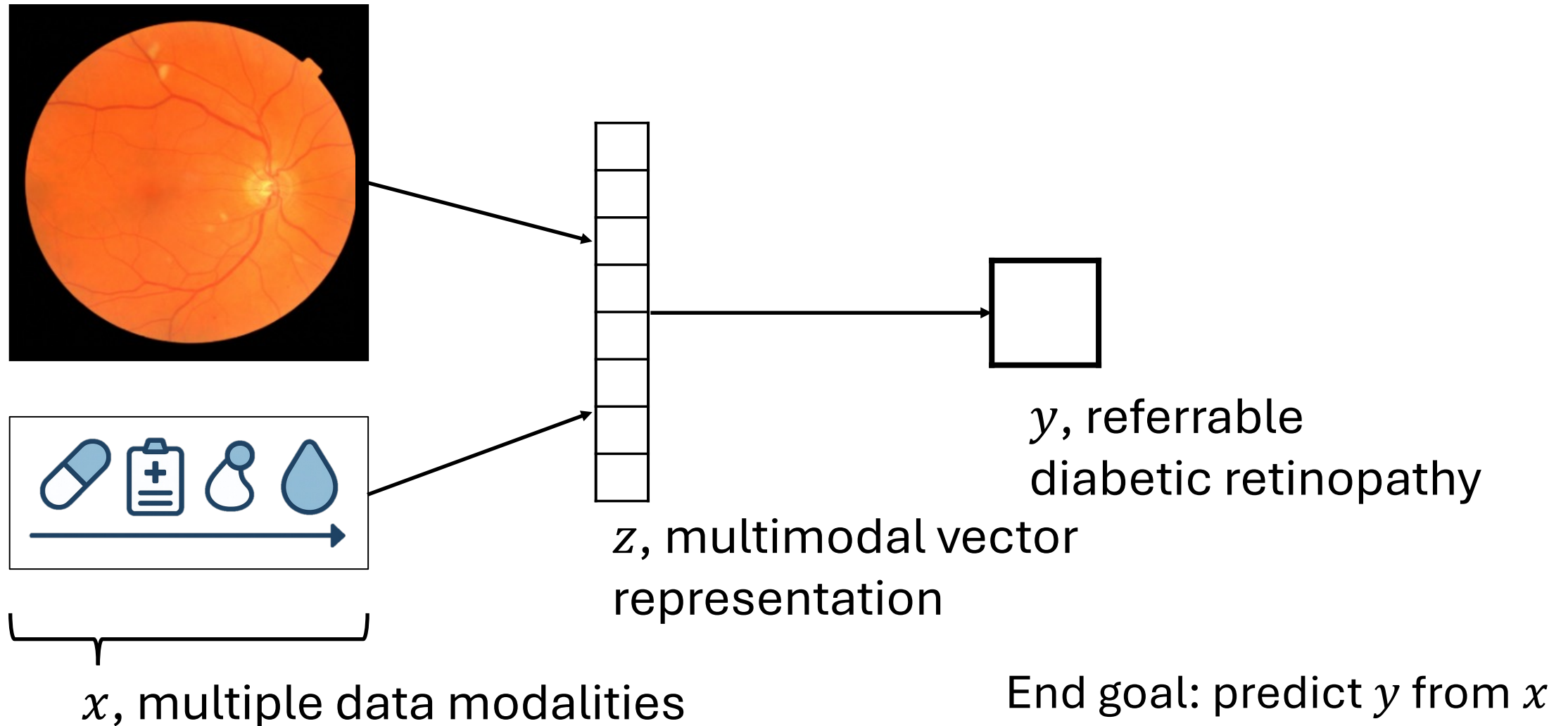
x , Go board state



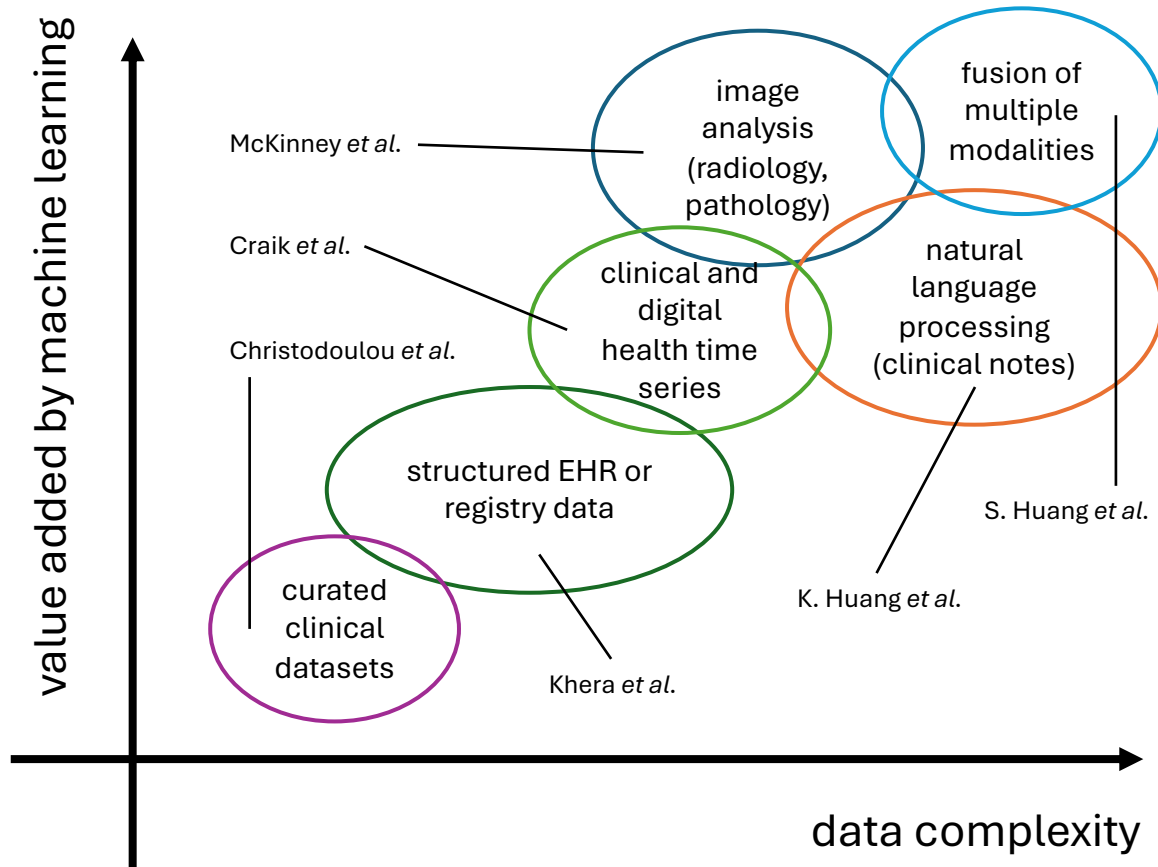
y , next move

End goal: predict y from x

Prediction from multimodal data: modality-specific encoders + multimodal fusion



As the complexity of the underlying data increases, so too does the probable value added by machine learning



Simple models based on clinical variables remain highly relevant

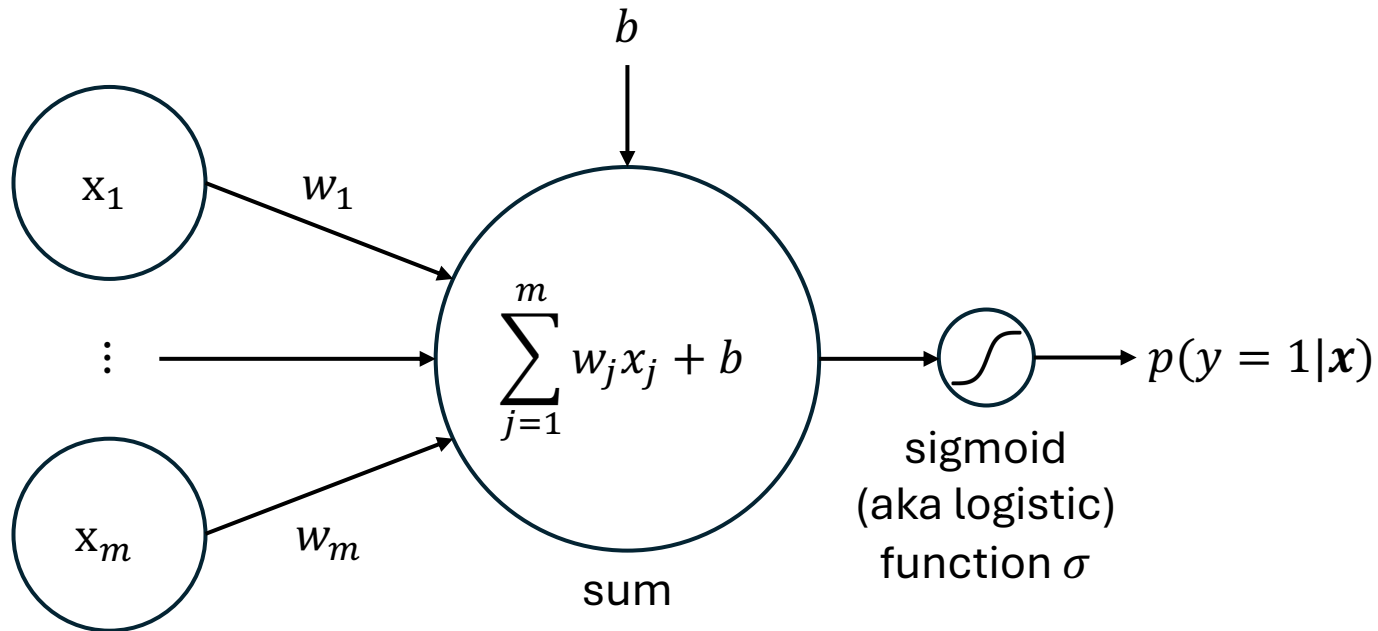
Use the simplest model that gets the job done

Engelhard MM, Navar AM, Pencina MJ. Incremental benefits of machine learning—when do we need a better mousetrap?. *JAMA cardiology*. 2021 Jun 1;6(6):621-3.

Logistic regression

A probabilistic perceptron

Logistic regression

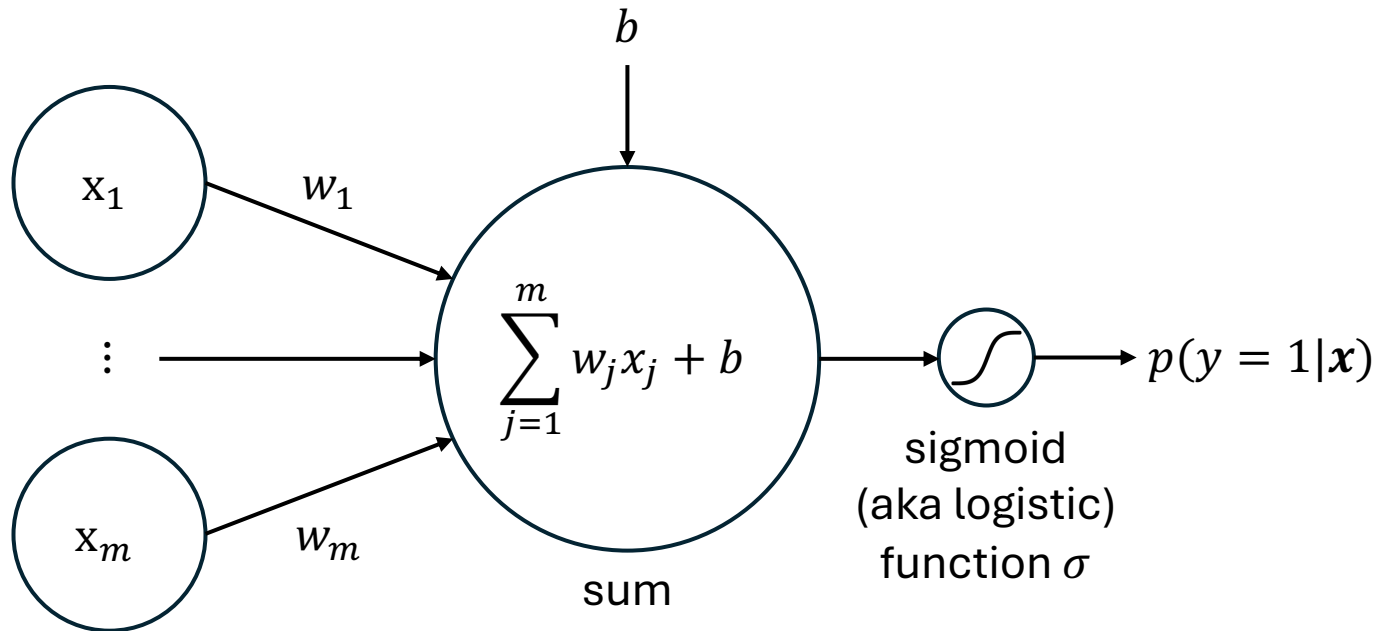


$$p(y = 1 | \mathbf{x}) = \sigma \left(\sum_{j=1}^m w_j x_j + b \right)$$

$$p(y = 1 | \mathbf{x}) = \sigma(\mathbf{w}^T \mathbf{x} + b)$$

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

Logistic regression



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$$p(y = 1|\mathbf{x}) = \sigma(\mathbf{w}^T \mathbf{x} + b)$$

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

(view 1) a generalized linear model with a logistic link function

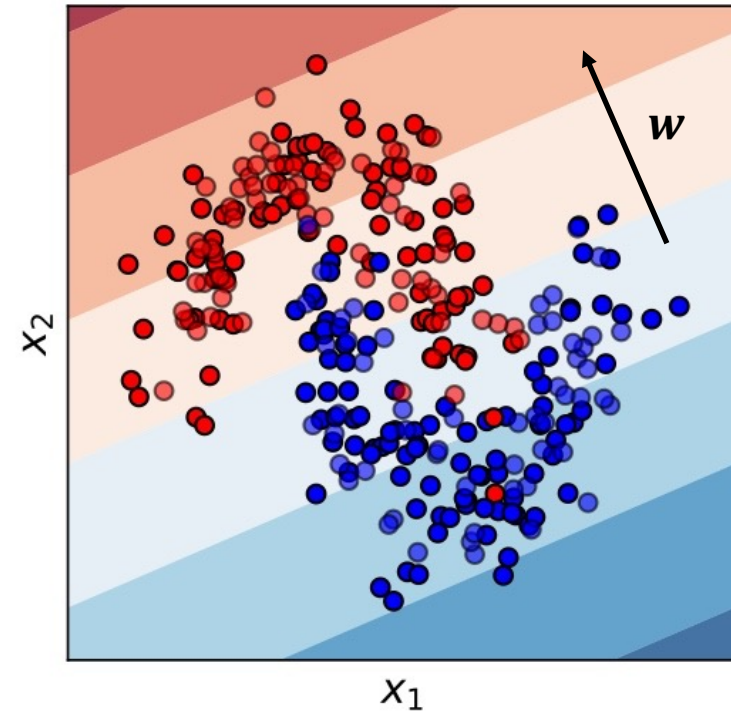
(view 2) a probabilistic perceptron, i.e. a single neuron of a neural network

Example: predict ICU mortality

- On admission, collect features $\mathbf{x}_i = (\text{age, sex, temp, BP, ...})$ and outcome y_i indicating whether patient i died during their ICU stay
- Using dataset $\mathcal{D} = \{\mathbf{x}_i, y_i\}_{i=1}^n$, find optimal parameters $\{\mathbf{w}, b\}$ (e.g., those that maximize the *likelihood* of $\mathcal{D}$; more on this later)
- Evaluate performance on a held-out dataset (more on this later)
- Use the model to predict $p(y = 1|\mathbf{x})$ for new patients that arrive

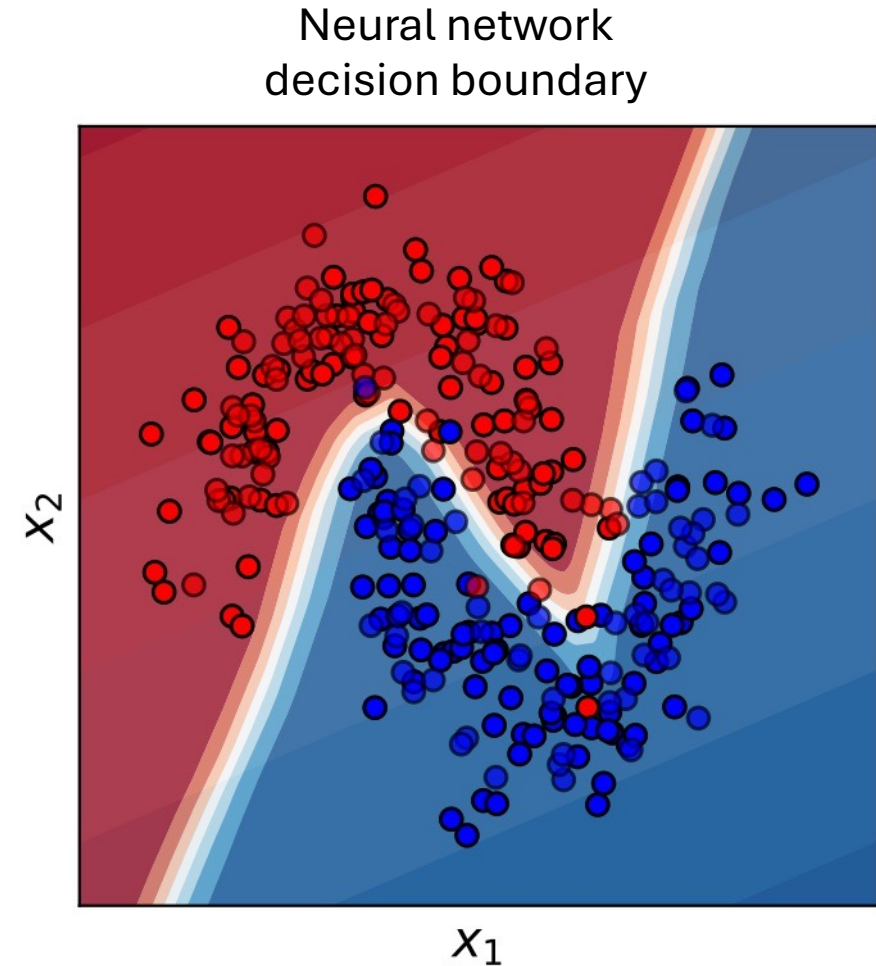
Logistic regression yields linear decision boundaries.

- Predicted log-odds increase as features move in the direction defined by $\mathbf{w}$
(consider the dot-product $\mathbf{w}^T \mathbf{x}$)
- Intercept b allows predicted log-odds to be non-zero at the origin
- Linear decision boundaries aren't always good enough



An impossible decision boundary for LR

- Suppose x_1 and x_2 are biomarker values
- After biopsy:
 - Blue patients: benign
 - Red patients: malignant
- Why can't logistic regression yield this decision boundary?
- Consider: how does increasing x_1 change the predicted log-odds?



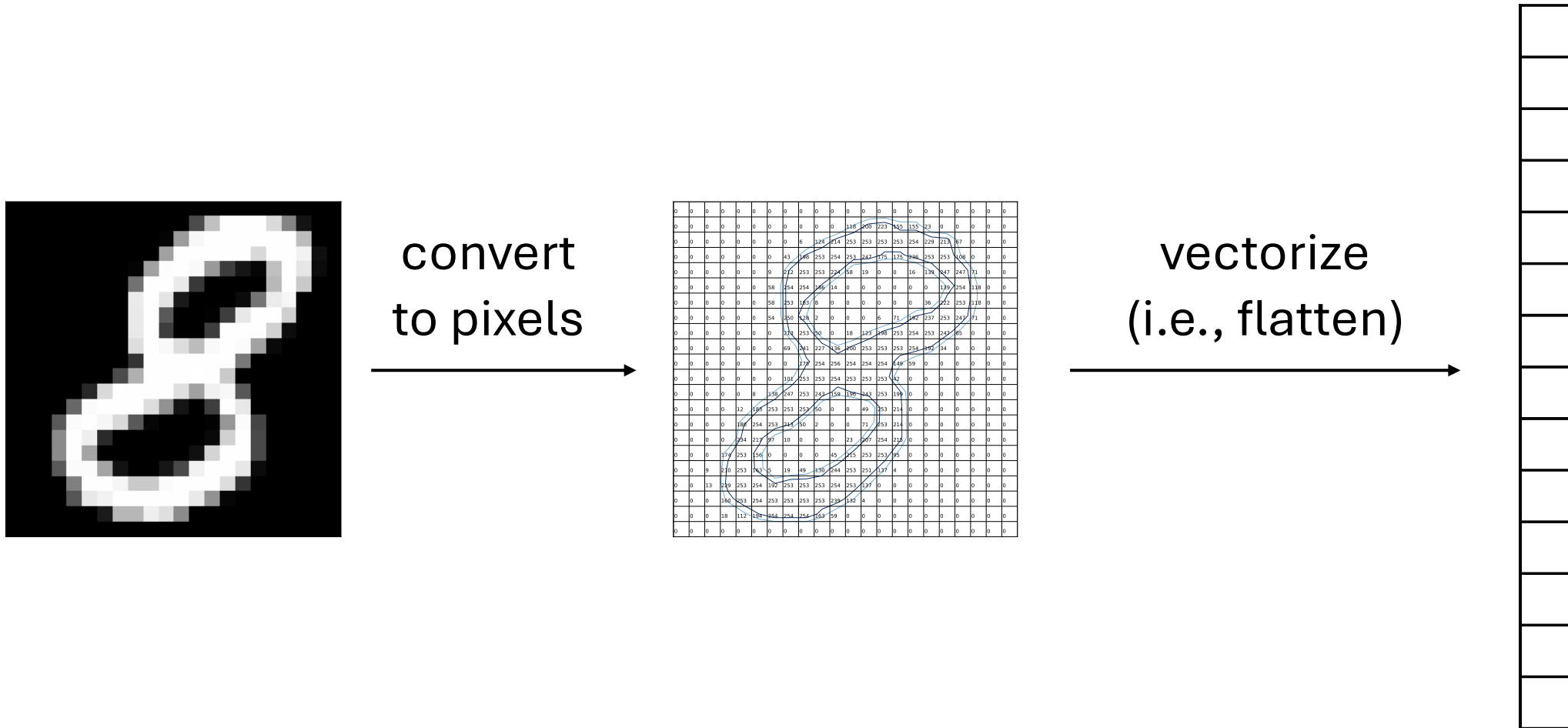
Recognizing Handwritten Digits

A visual example

MNIST: Identify handwritten digits

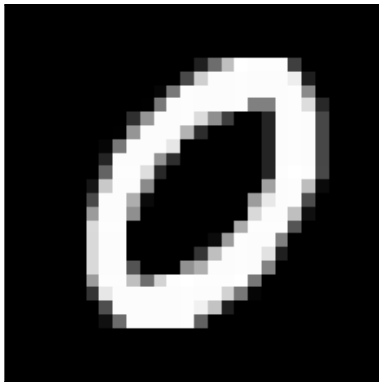
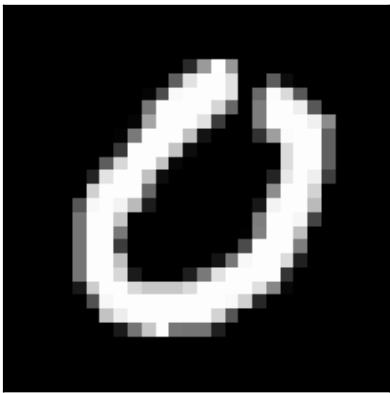
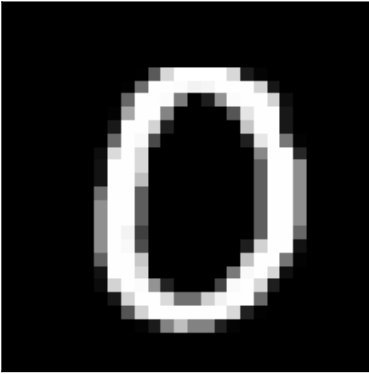


To use logistic regression, we must convert pixels to numbers, then vectorize

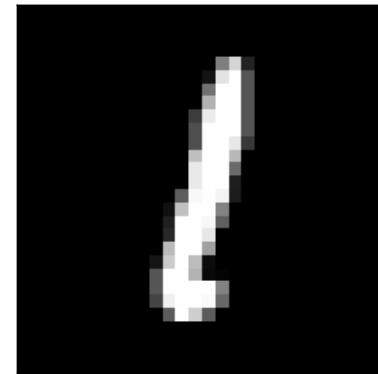
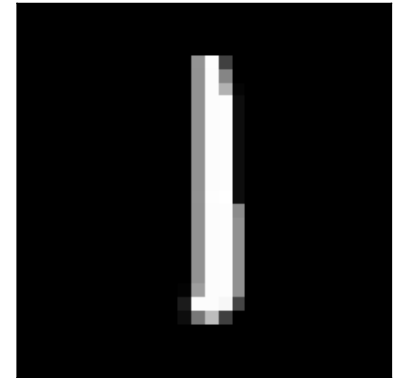
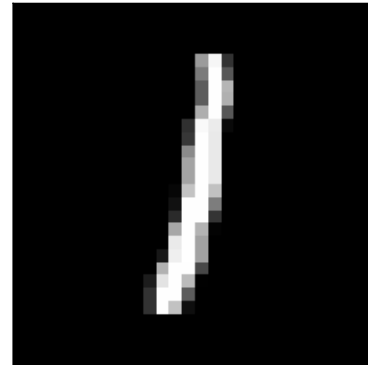


Start With The Binary Case

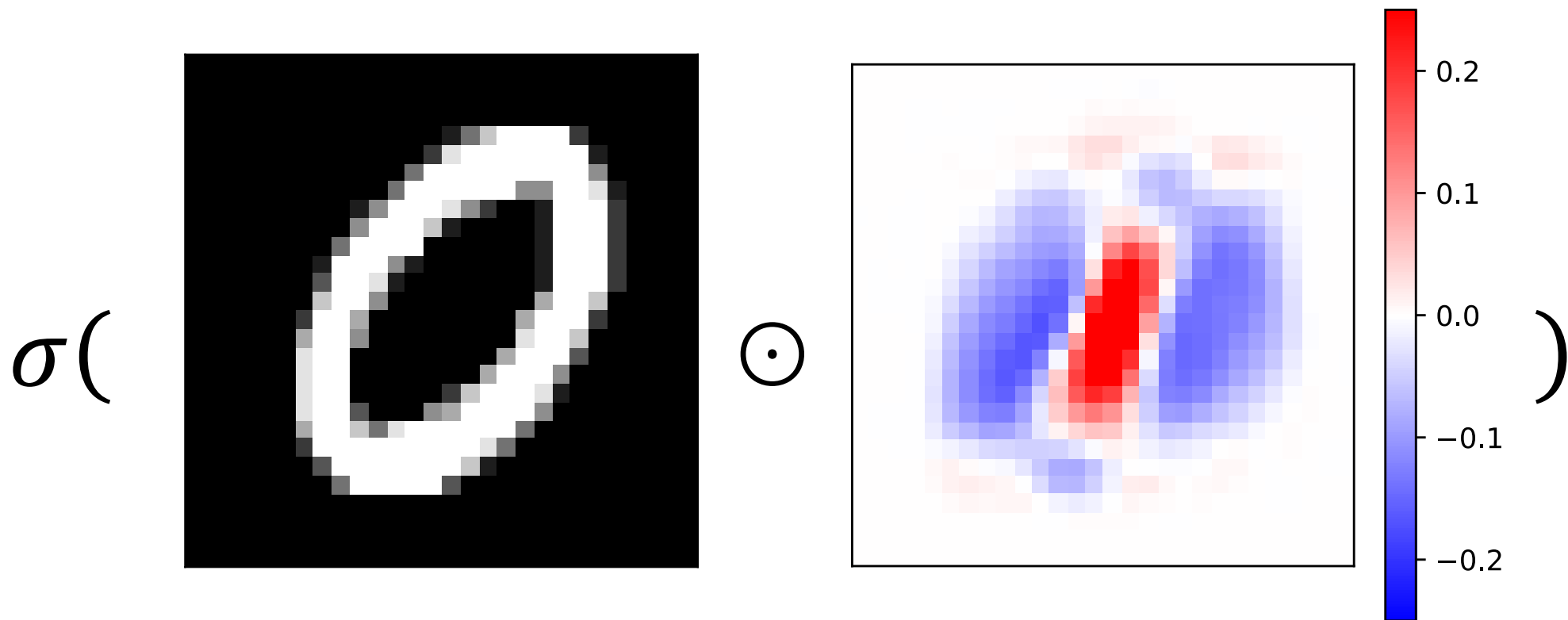
Zeros



Ones

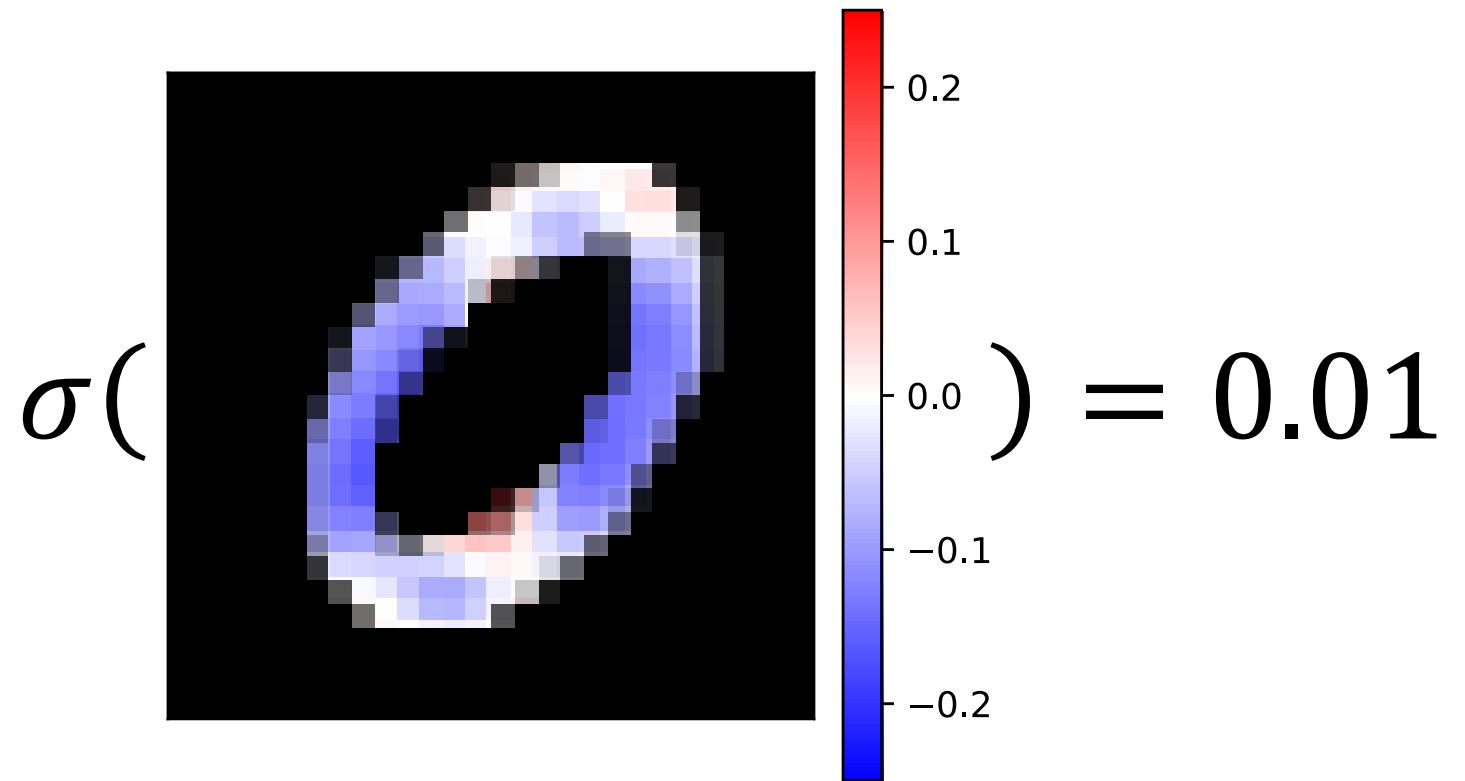


We may view the learned w as a *filter*



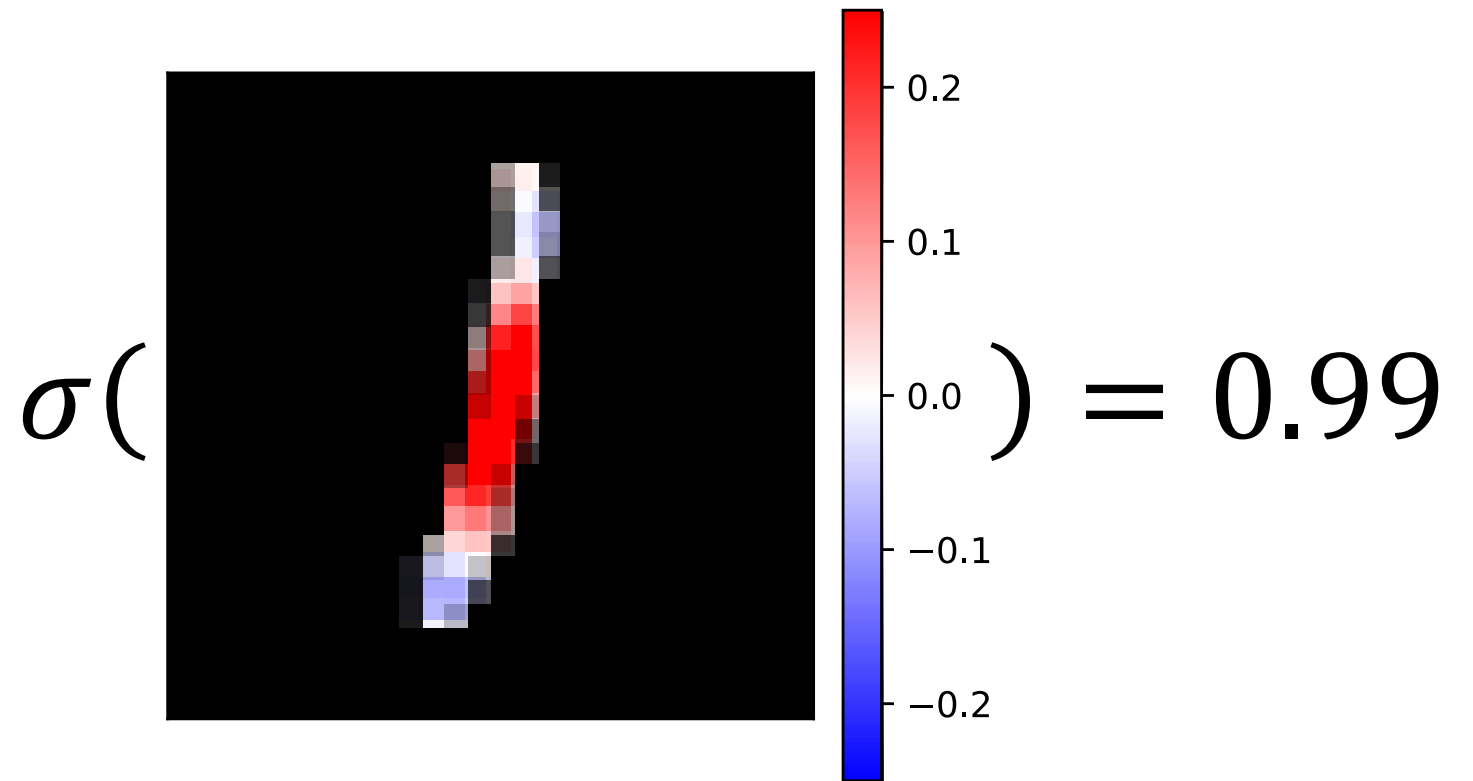
We may view the learned w as a *filter*

- Superimpose and sum to estimate the dot product
- When **blue** regions are revealed, predicted log-odds are negative



We may view the learned w as a *filter*

- Superimpose and sum to estimate the dot product
- When **red** regions are revealed, predicted log-odds are positive



Logistic regression: *classification, or risk prediction?*

Is logistic regression a *classification* algorithm?

(discuss and report back in 2 minutes)

Logistic regression: *classification, or risk prediction?*

- **Classification**

- Examples: MNIST, EHR-based phenotyping, pathology detection
- Labels are deterministic
- Positives and negatives have non-overlapping support (in principle, given the right features)

- **Risk Prediction**

- Examples: ICU mortality, heart attack risk, future diagnosis likelihood
- Labels are inherently probabilistic
- Positives and negatives have overlapping support in any feature space

Summary and what's next

- In this class, we will learn how to build neural network prediction machines from tabular, image, text, and multimodal data.
- Logistic regression is an important prediction machine, and it can be viewed as the building block of the neural network. For these reasons, it is important that you understand it well.
- Next time, we will focus on building neural networks that overcome the key limitation of logistic regression, namely its linear decision boundary.
- Then, we will focus on model learning via stochastic gradient methods.